

Extensions of PerfectGroup(9720,2) via $Z\phi$ -classes

zphi method, Claude Archer

Solvable quotient groups of orders $2 \leq n \leq 127$

Here $|S| = 2n$ and $|E| = 9720n$. GAP runtimes are reported in seconds.

Let $P = \text{PerfectGroup}(9720, 2)$. In this computation one has

$$Z(P) \simeq C_2, \quad \text{Out}(P) \simeq C_6, \quad P/Z(P) \simeq \text{PerfectGroup}(4860, 2).$$

Since $\text{Aut}(C_2) = 1$, there is no marking ambiguity and no non-trivial central action to record. The auxiliary group S has order $2n$ and contains a distinguished central involution z identified with $Z(P)$. For each orbit of such central involutions under $\text{Aut}(S)$, the quotient

$$H = S/\langle z \rangle$$

is tested for all couplings

$$\phi : H \longrightarrow \text{Out}(P) \simeq C_6.$$

The computation is not restricted to surjective couplings. The possible images are exactly

$$1, \quad C_2, \quad C_3, \quad C_6,$$

and they are displayed below as the columns *Central*, C_2 , C_3 and C_6 .

How the couplings are classified. Fix a central involution $z \in Z(S)$ and put $H = S/\langle z \rangle$. A coupling with image $U \leq C_6$ is determined by a normal subgroup $K \trianglelefteq H$ with $H/K \simeq U$, together with an identification $H/K \rightarrow U$. Two such identifications may define equivalent $Z\phi$ -data if they differ by an automorphism induced from the stabilizer of z in $\text{Aut}(S)$. Equivalently, for each kernel K , one counts the corresponding double-coset quotient inside $\text{Aut}(U)$, as in the GrpConst-style classification of couplings. Since C_6 is abelian, the normalizer action of $N_{C_6}(U)$ on U is trivial; thus no additional conjugation inside $\text{Out}(P)$ identifies the different embeddings. This explains why the non-surjective couplings with images C_2 and C_3 must be counted separately from the surjective image C_6 .

Table of $Z\phi$ -classes

The following tables use almost the full available text width. The columns *Cumulative* and *Cum. time* are global cumulative sums from $n = 2$ onward.

Values for $2 \leq n \leq 17$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
2	4	19 440	2	2	0	0	4	4	0.337	0.337
3	6	29 160	1	0	1	0	2	6	0.085	0.422
4	8	38 880	6	8	0	0	14	20	0.558	0.980
5	10	48 600	1	0	0	0	1	21	0.058	1.038
6	12	58 320	4	4	2	2	12	33	6.965	8.003
7	14	68 040	1	0	0	0	1	34	0.023	8.026
8	16	77 760	21	42	0	0	63	97	0.745	8.771
9	18	87 480	2	0	2	0	4	101	0.064	8.835
10	20	97 200	4	4	0	0	8	109	0.439	9.274
11	22	106 920	1	0	0	0	1	110	0.021	9.295
12	24	116 640	16	24	8	8	56	166	1.303	10.598
13	26	126 360	1	0	0	0	1	167	0.018	10.616
14	28	136 080	4	4	0	0	8	175	0.199	10.815
15	30	145 800	1	0	1	0	2	177	0.028	10.843
16	32	155 520	95	286	0	0	381	558	6.932	17.775
17	34	165 240	1	0	0	0	1	559	0.020	17.795

Values for $18 \leq n \leq 33$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
18	36	174 960	10	10	6	6	32	591	0.954	18.749
19	38	184 680	1	0	0	0	1	592	0.024	18.773
20	40	194 400	16	26	0	0	42	634	1.251	20.024
21	42	204 120	2	0	3	0	5	639	0.150	20.174
22	44	213 840	4	4	0	0	8	647	0.227	20.401
23	46	223 560	1	0	0	0	1	648	0.029	20.430
24	48	233 280	72	166	26	46	310	958	9.020	29.450
25	50	243 000	2	0	0	0	2	960	0.039	29.489
26	52	252 720	4	4	0	0	8	968	0.310	29.799
27	54	262 440	5	0	8	0	13	981	0.447	30.246
28	56	272 160	14	24	0	0	38	1 019	1.518	31.764
29	58	281 880	1	0	0	0	1	1 020	0.021	31.785
30	60	291 600	8	8	4	4	24	1 044	1.744	33.529
31	62	301 320	1	0	0	0	1	1 045	0.133	33.662
32	64	311 040	649	3 026	0	0	3 675	4 720	233.986	267.648
33	66	320 760	1	0	1	0	2	4 722	0.030	267.678

Values for $34 \leq n \leq 49$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
34	68	330 480	4	4	0	0	8	4 730	0.307	267.985
35	70	340 200	1	0	0	0	1	4 731	0.026	268.011
36	72	349 920	48	80	26	32	186	4 917	7.855	275.866
37	74	359 640	1	0	0	0	1	4 918	0.017	275.883
38	76	369 360	4	4	0	0	8	4 926	0.199	276.082
39	78	379 080	2	0	3	0	5	4 931	0.133	276.215
40	80	388 800	73	176	0	0	249	5 180	15.513	291.728
41	82	398 520	1	0	0	0	1	5 181	0.018	291.746
42	84	408 240	12	12	12	12	48	5 229	2.016	293.762
43	86	417 960	1	0	0	0	1	5 230	0.049	293.811
44	88	427 680	14	24	0	0	38	5 268	1.637	295.448
45	90	437 400	2	0	2	0	4	5 272	0.095	295.543
46	92	447 120	4	4	0	0	8	5 280	0.224	295.767
47	94	456 840	1	0	0	0	1	5 281	0.018	295.785
48	96	466 560	422	1 574	116	306	2 418	7 699	107.288	403.073
49	98	476 280	2	0	0	0	2	7 701	0.042	403.115

Values for $50 \leq n \leq 64$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
50	100	486 000	10	10	0	0	20	7 721	1.047	404.162
51	102	495 720	1	0	1	0	2	7 723	0.058	404.220
52	104	505 440	16	26	0	0	42	7 765	1.525	405.745
53	106	515 160	1	0	0	0	1	7 766	0.019	405.764
54	108	524 880	30	30	30	30	120	7 886	6.190	411.954
55	110	534 600	2	0	0	0	2	7 888	0.159	412.113
56	112	544 320	64	158	0	0	222	8 110	9.735	421.848
57	114	554 040	2	0	3	0	5	8 115	0.183	422.031
58	116	563 760	4	4	0	0	8	8 123	0.297	422.328
59	118	573 480	1	0	0	0	1	8 124	0.098	422.426
60	120	583 200	46	84	18	26	174	8 298	15.909	438.335
61	122	592 920	1	0	0	0	1	8 299	0.027	438.362
62	124	602 640	4	4	0	0	8	8 307	0.264	438.626
63	126	612 360	4	0	7	0	11	8 318	0.617	439.243
64	128	622 080	8 308	72 010	0	0	80 318	88 636	2268.448	2707.691

Values for $65 \leq n \leq 80$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
65	130	631 800	1	0	0	0	1	88 637	0.000	2707.691
66	132	641 520	8	8	4	4	24	88 661	4.208	2711.899
67	134	651 240	1	0	0	0	1	88 662	0.000	2711.899
68	136	660 960	16	26	0	0	42	88 704	4.271	2716.170
69	138	670 680	1	0	1	0	2	88 706	0.209	2716.379
70	140	680 400	8	8	0	0	16	88 722	2.354	2718.733
71	142	690 120	1	0	0	0	1	88 723	0.000	2718.733
72	144	699 840	262	655	107	220	1 244	89 967	219.076	2937.809
73	146	709 560	1	0	0	0	1	89 968	0.000	2937.809
74	148	719 280	4	4	0	0	8	89 976	0.796	2938.605
75	150	729 000	3	0	3	0	6	89 982	0.442	2939.047
76	152	738 720	14	24	0	0	38	90 020	3.826	2942.873
77	154	748 440	1	0	0	0	1	90 021	0.000	2942.873
78	156	758 160	12	12	12	12	48	90 069	7.615	2950.488
79	158	767 880	1	0	0	0	1	90 070	0.000	2950.488
80	160	777 600	445	1 664	0	0	2 109	92 179	158.054	3108.542

Values for $81 \leq n \leq 96$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
81	162	787 320	15	0	34	0	49	92 228	2.457	3110.999
82	164	797 040	4	4	0	0	8	92 236	0.807	3111.806
83	166	806 760	1	0	0	0	1	92 237	0.000	3111.806
84	168	816 480	56	104	48	72	280	92 517	51.114	3162.920
85	170	826 200	1	0	0	0	1	92 518	0.000	3162.920
86	172	835 920	4	4	0	0	8	92 526	0.786	3163.706
87	174	845 640	1	0	1	0	2	92 528	0.212	3163.918
88	176	855 360	63	158	0	0	221	92 749	18.874	3182.792
89	178	865 080	1	0	0	0	1	92 750	0.000	3182.792
90	180	874 800	20	20	12	12	64	92 814	13.445	3196.237
91	182	884 520	1	0	0	0	1	92 815	0.000	3196.237
92	184	894 240	14	24	0	0	38	92 853	2.823	3199.060
93	186	903 960	2	0	3	0	5	92 858	0.280	3199.340
94	188	913 680	4	4	0	0	8	92 866	0.780	3200.120
95	190	923 400	1	0	0	0	1	92 867	0.000	3200.120
96	192	933 120	3 914	23 798	752	3 182	31 646	124 513	4426.289	7626.409

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
97	194	942 840	1	0	0	0	1	124 514	0.000	7626.409
98	196	952 560	10	10	0	0	20	124 534	2.699	7629.108
99	198	962 280	2	0	2	0	4	124 538	0.365	7629.473
100	200	972 000	52	88	0	0	140	124 678	15.862	7645.335
101	202	981 720	1	0	0	0	1	124 679	0.000	7645.335
102	204	991 440	8	8	4	4	24	124 703	5.142	7650.477
103	206	1 001 160	1	0	0	0	1	124 704	0.000	7650.477
104	208	1 010 880	73	176	0	0	249	124 953	25.946	7676.423
105	210	1 020 600	2	0	3	0	5	124 958	0.471	7676.894
106	212	1 030 320	4	4	0	0	8	124 966	0.986	7677.880
107	214	1 040 040	1	0	0	0	1	124 967	0.000	7677.880
108	216	1 049 760	162	288	148	192	790	125 757	180.261	7858.141
109	218	1 059 480	1	0	0	0	1	125 758	0.000	7858.141
110	220	1 069 200	12	12	0	0	24	125 782	3.651	7861.792
111	222	1 078 920	2	0	3	0	5	125 787	0.416	7862.208
112	224	1 088 640	383	1 520	0	0	1 903	127 690	138.494	8000.702

Values for $113 \leq n \leq 127$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
113	226	1 098 360	1	0	0	0	1	127 691	0.000	8000.702
114	228	1 108 080	12	12	12	12	48	127 739	7.953	8008.655
115	230	1 117 800	1	0	0	0	1	127 740	0.000	8008.655
116	232	1 127 520	16	26	0	0	42	127 782	3.762	8012.417
117	234	1 137 240	4	0	7	0	11	127 793	0.680	8013.097
118	236	1 146 960	4	4	0	0	8	127 801	1.011	8014.108
119	238	1 156 680	1	0	0	0	1	127 802	0.000	8014.108
120	240	1 166 400	275	760	82	184	1 301	129 103	289.103	8303.211
121	242	1 176 120	2	0	0	0	2	129 105	0.000	8303.211
122	244	1 185 840	4	4	0	0	8	129 113	1.073	8304.284
123	246	1 195 560	1	0	1	0	2	129 115	0.334	8304.618
124	248	1 205 280	14	24	0	0	38	129 153	3.442	8308.060
125	250	1 215 000	5	0	0	0	5	129 158	0.000	8308.060
126	252	1 224 720	32	32	40	40	144	129 302	26.572	8334.632
127	254	1 234 440	1	0	0	0	1	129 303	0.000	8334.632
Range		Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)	
$2 \leq n \leq 127$		16 007	107 331	1 559	4 406	129 303	129 303	8334.632	8334.632	

Largest contributors to the total runtime

The total recorded GAP runtime for $2 \leq n \leq 127$ is

$$8334.632 \text{ s} = 138.911 \text{ min} \simeq 2 \text{ h } 18 \text{ min } 55 \text{ s}.$$

For $n = 64$, the $Z\phi$ -class counts in the new run agree exactly with the previously tabulated row:

$$(8308, 72010, 0, 0), \quad \#Z\phi = 80318.$$

The runtime at $n = 64$ has been updated to the value measured in the new $n = 64, \dots, 127$ run; runtimes may vary between runs and do not affect the class counts. The following values of n dominate the recorded computation. Times are shown both in seconds and in minutes.

n	Time (s)	Time (min)	$\#Z\phi$	Runtime share	Class share
96	4426.289	73.771	31 646	53.11%	24.47%
64	2268.448	37.807	80 318	27.22%	62.12%
120	289.103	4.818	1 301	3.47%	1.01%
32	233.986	3.900	3 675	2.81%	2.84%
72	219.076	3.651	1 244	2.63%	0.96%
108	180.261	3.004	790	2.16%	0.61%
80	158.054	2.634	2 109	1.90%	1.63%
112	138.494	2.308	1 903	1.66%	1.47%

The two largest computations are $n = 96$ and $n = 64$. Together they account for about 80.32% of the recorded runtime and 86.59% of all computed $Z\phi$ -classes. As in the smaller range, values with a large 2-part tend to dominate the enumeration because they produce many auxiliary groups S and many central-involution orbits.